

Course Description

Prof. Dr. Gerardo Flores

Microprocessor Systems (CSCE 3340 - SENG 3345)

TAMIU



Instructor Information

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To schedule an appointment, please contact me through Blackboard or email.

Microprocessor Systems (CSCE 3340 - SENG 3345)

Course Description

Basic computer structure, the instruction set, addressing modes, **assembly language programming**, assembly language subroutines, arithmetic operations, **programming in C**, implementation of C procedures, elementary data structures, **input and output** and a survey of microprocessor based design. This course is interchangeable with SENG 3345.

Prerequisite: CSCE 2330 Digital Logic Design

Additional Course Information:

- Introduction to microprocessor systems with **assembly language** and **C programming**.
- Focus on **microprocessor architecture**, peripheral programming, and **hands-on practice**.
- Use of simulation tools: **Cpulator**, **Keil µVision**, **STM32CubeIDE**, and **Wokwi**.
- Quick review of basic concepts to work with **ARM Cortex microcontrollers**.
- AI tools may be used for learning and debugging unless otherwise stated. Students remain responsible for understanding and explaining all submitted work. AI tools are not permitted during examinations.

Program Learning Outcomes

Foundational Concepts:

- Understand differences between microprocessors and microcontrollers.
- Explore Harvard vs. Von Neumann architectures.
- Proficiency in data representation (binary, hexadecimal, signed/unsigned integers).

Assembly and C Programming:

- Write, debug, and integrate assembly and C programs for ARM Cortex-M microcontrollers.
- Practice arithmetic, logical operations, control flow, and memory access.

Simulation and Tools:

- Use Cputator, Keil μ Vision and Wokwi to simulate and test functionality like **GPIO, ADC, PWM, and interrupts**.

Hardware-Software Integration:

- Configure and program peripherals (GPIO, timers, ADC, PWM).
- Manage interrupts (NVIC, ISRs).

Embedded System Design:

- Design, simulate, and implement embedded systems.
- Apply debugging and problem-solving techniques.
- Complete a project combining theoretical knowledge with real-world applications.

Student Learning Outcomes

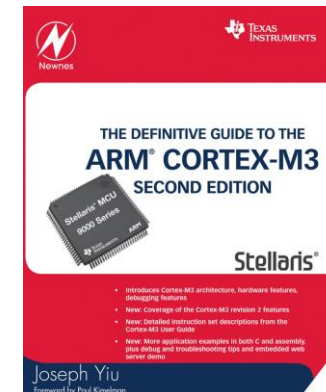
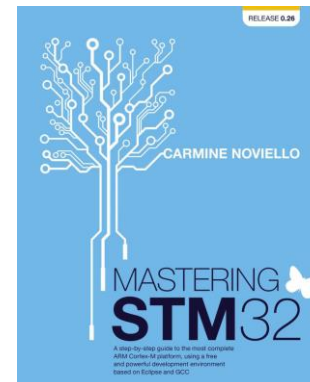
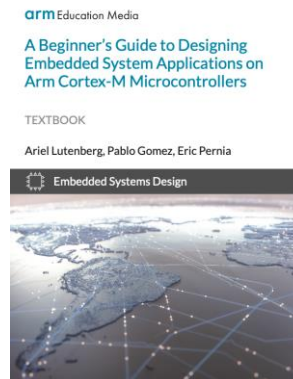
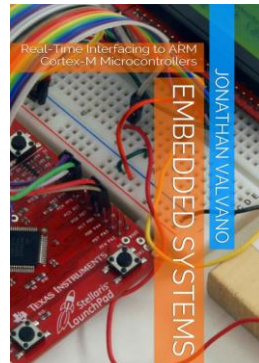
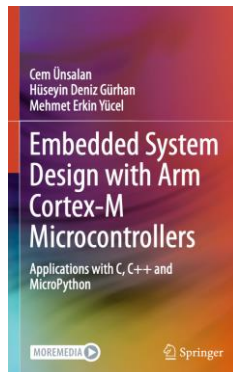
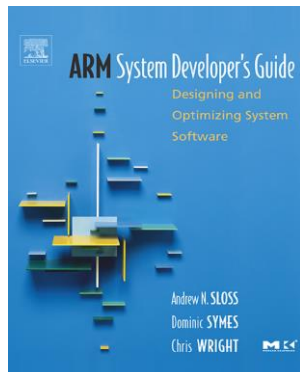
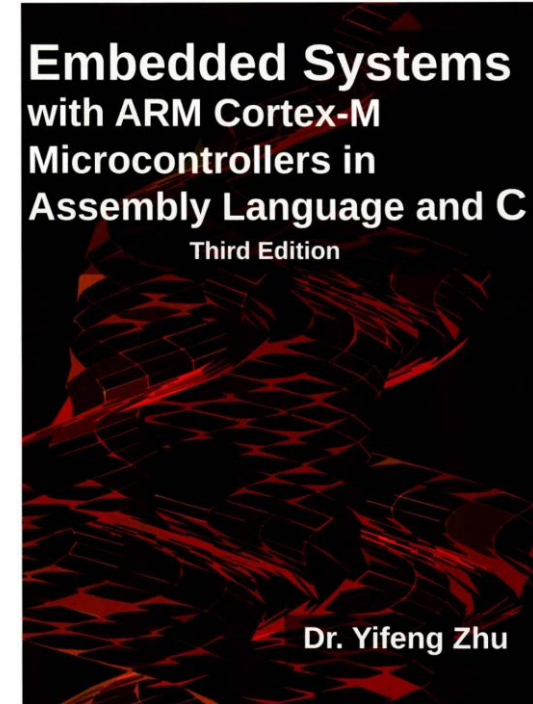
1. Develop a solid understanding of microprocessor architecture and programming, including assembly language and C for **ARM Cortex-M microcontrollers**.
2. Gain proficiency in using industry-standard tools like **Cpulator** and **Wokwi** to simulate, program, and debug embedded systems.
3. Design and analyze embedded systems for various applications, focusing on peripheral integration, interrupts, and hardware-software interaction.

Important Dates

Visit the Academic Calendar (tamiu.edu) (<https://www.tamiu.edu/academiccalendar/>) page to view the term's important dates.

Textbooks: No textbook is required. The following books are recommended references.

Group	Title	Author	ISBN
Optional	Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C (Third Edition)	Yifeng Zhu	9780982692660
Optional	The Definitive Guide to the ARM Cortex-M3 (Second Edition)	Joseph Yiu	978-1-85617-963-8
Optional	ARM System Developer's Guide: Designing and Optimizing System Software	Andrew N. Sloss, Dominic Symes, Chris Wright	978-0121477513
Optional	Embedded System Design with ARM Cortex-M Microcontrollers	Cem Ünsalan, Hüseyin Deniz Gürhan, Mehmet Erkin Yücel	978-3-030-88438-3
Optional	A Beginner's Guide to Designing Embedded System Applications on ARM Cortex-M Microcontrollers	Ariel Lutenberg, Pablo Gomez, Eric Pernia	9781911531418
Optional	Embedded Systems: Real-Time Interfacing to Arm Cortex-M Microcontrollers	Jonathan Valvano	978-1463590154
Optional	Mastering STM32	Carmine Noviello	NA



Other Course Materials

1. Cputator Simulator: For developing and debugging assembly and C programs for microcontrollers.
2. Wokwi Simulator: An online platform for simulating embedded systems, allowing students to program in C and visualize peripherals such as GPIO, ADC, and PWM.
3. STM32CubeIDE (optional): An integrated development environment for STM32 microcontrollers, supporting both C programming and debugging features.

Additionally, students will have access to notes, reference materials, and documentation for ARM Cortex-M microcontrollers. These resources will provide both theoretical and practical insights into microprocessor architecture, programming, and real-world applications.

Students are encouraged to install and familiarize themselves with these tools before the start of the course to maximize hands-on learning.

Grading Criteria

- Assignments – 40%
- Midterm Exam – 30%
- Final Exam – 30%

Grading Criteria

GRADE	PERCENTAGE
A	91-100
B	80-90.9
C	70-79.9
D	60-69.9
F	Below 60

Assignments

- Tasks include readings, research, exercises, and coding challenges aligned with lecture topics.
- Designed to reinforce understanding and provide hands-on experience with theoretical concepts.
- Some assignments may offer **extra credit**, evaluated on complexity and effort, to boost final grades.

Course Contents

1. Microprocessors and Microcontrollers
2. Harvard and Von Neumann Architectures
3. Data Representation and Number Systems
4. ARM Cortex-M Architecture and Instruction Set
- 5. Assembly Language Programming**
6. Arithmetic, Logical, and Bitwise Operations
7. Memory Access and Addressing Modes
8. Control Flow and Conditional Branching
9. Subroutines and Stack Management
- 10. C Programming** and Assembly–C Integration
11. Interrupts and NVIC Configuration
12. Digital Input and Output (GPIO)
13. Timers and Pulse-Width Modulation (PWM)
14. Analog-to-Digital Conversion (ADC)
15. Embedded-System Design and Applications

Note: A detailed tentative schedule, including dates and assignments, will be published in the course syllabus on Blackboard this week.

Course Communication and Materials

- Please check Blackboard regularly. All course announcements, grades, and course content will be posted there.
- Code, slides, and other supplementary materials will be available at:
- <https://github.com/gfloresc/Microprocessor-Systems/tree/main>
- The GitHub link will also be posted on Blackboard.
- The course syllabus will be updated this week.

Any Questions?

Thank you for your attention!

We will begin the course content in our next class session.